Protocol – Preparation of Bacterial Crude Extract

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

This protocol lyses the cell pellets from Protocol 4 enzymatically with lysozyme and DNase I and yields the soluble fraction, the cell-free extract (CFE), which should contain the expressed enzyme. Volumes are given for about 3.5 mL of cell suspension per tube.

Materials list

- Frozen cell pellets
- **50 mM potassium phosphate buffer (KH_2PO_4), pH 7.5**
- **Lysozyme** (working solution 17.5 mg/mL; **0.5 mg/mL final**)
- **DNase I** (working solution about 350 µg/mL; **10 µg/mL final**)
- **$MgCl_2$, 0.5 M stock** (**2 mM final**)
- Vortex mixer
- Thermoshaker or orbital shaker (room temperature, **140 rpm**)
- Refrigerated centrifuge (**10 °C**)
- Micropipettes, sterile tips, vials, ice
- **–80 °C** freezer (for storage of unused CFE)

Steps

1. Thaw the frozen pellets and resuspend each in 50 mM potassium phosphate buffer (KH_2PO_4), pH 7.5, at about **3.5 mL per pellet**. Vortex until homogeneous.
2. Add lysozyme to a final concentration of **0.5 mg/mL** (for a 3.5 mL suspension, **100 µL of a 17.5 mg/mL** lysozyme solution) to digest the cell wall.
3. Add DNase I to a final concentration of **10 µg/mL** (**100 µL of a 350 µg/mL** DNase I solution per tube) to degrade the released genomic DNA.
4. Add $MgCl_2$ to a final concentration of **2 mM** (**14 µL of a 0.5 M** $MgCl_2$ solution per tube) as the DNase I cofactor.
5. Agitate at **room temperature, 140 rpm, for 30 min** on the thermoshaker. The suspension clears as the cells lyse.
6. Centrifuge at **8500 rpm, 10 °C, for 10 min**.
7. Collect the supernatant, the cell-free extract (CFE), with a pipette and keep it **on ice**.
8. Store any unused CFE at **–80 °C** in the same vials.

Troubleshooting

- **Turbid extract:** Nucleic-acid fragments can be removed by adding **streptomycin sulfate to 1% (w/v)** and centrifuging again. In our case the extract was clear, so both this step and mechanical sonication were omitted.

- **Incorrect centrifugation conditions:** Step 6 specifies centrifugation in **rpm rather than ×g**. Because rpm depends on rotor radius, the corresponding relative centrifugal force should be determined and reported.
- **Loss of activity on storage:** Store the CFE in single-use aliquots to avoid repeated freeze–thaw cycles.